

Project Executable Files

Date	4 July 2026
Team ID	xxxxxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
EduGenie-Google-Gemini-Powered-Learning-Assistant/
|
|-- app/                                # FastAPI Backend Server
|   |-- main.py                        # API Gateway Entrypoint
|   |-- config.py                     # Port & Path Settings
|   |-- models/                       # Pydantic Schemas
|       |-- schemas.py
|   |-- routers/                      # API Endpoints Router
|       |-- learning.py               # explain/chat/quiz endpoints
|   |-- services/                     # Business Logic Layers
|       |-- gemini_service.py
|       |-- history_logger.py
|       |-- feedback_logger.py
|       |-- pdf_generator.py
|
|-- frontend/                          # Streamlit Web UI Interface
|   |-- streamlit_app.py              # Main Streamlit Application
|   |-- assets/                       # Embedded Logos & Graphics
|       |-- index.html                # HTML Layout Template
|       |-- edugenie_logo.png
|       |-- robot_workspace.png
|       |-- sidebar_decoration.png
|
|-- tests/                             # QA Automated Testing Suite
|   |-- test_backend.py               # Pytest router test cases
|
|-- README.md                         # Documentation & Run Manual
|-- package.json                     # Node.js project configs
|-- run_backend.bat                   # FastAPI Startup Script
|-- run_frontend.bat                 # Streamlit Startup Script
|-- run_edugenie.bat                 # Unified Orchestration Script
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not yet deployed — runs locally at http://localhost:8501 (frontend) with the API at http://localhost:8000 .
Login Credentials (Demo)	No pre-seeded demo account exists; the app operates with open access, allowing immediate topic search and quiz generation.
Platform / Hosting Provider	Not yet hosted. Render (or AWS EC2 / Elastic Beanstalk) is the proposed hosting approach documented in the Technology Stack section — not yet implemented.
Repository Link	https://github.com/manankochar1720/Gen-ai-Google-Gemini-Powered-Learning-Assistant
Demo Video Link	https://drive.google.com/file/d/1bKsPvajDTE8vF-AaRfxUXIssAQp171tJ/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

- Python 3.11+
- Node.js 18+ and npm 9+ (optional)
- Git
- SQLite (comes pre-installed with Python)
- Google Gemini API Key

Step 1: Clone the repository:

```
git clone https://github.com/[INSERT_YOUR_GITHUB_REPO_LINK_HERE]
```

Step 2: Navigate to the project directory:

```
cd "EduGenie Google Gemini Powered Learning Assistant"
```

Step 3: Setup and run the backend (FastAPI):

- Create virtual environment: `python -m venv venv`
- Activate env: `venv\Scripts\activate` (Windows) or `source venv/bin/activate` (Mac/Linux)
- Install dependencies: `pip install -r requirements.txt` (or run `npm install` inside root)
- Set API Key in `.env`: `GEMINI_API_KEY=your_gemini_key_here`
- Start Uvicorn: `run_backend.bat` or `python -m uvicorn app.main:app --port 8000 --reload`

Step 4: Setup and run the frontend (Streamlit):

- Open a new terminal window in the root directory.
- Start UI: `run_frontend.bat` or `streamlit run frontend/streamlit_app.py`

Step 5: Open your browser and visit: `http://localhost:8501`

Step 6: Register or Login on the frontend page to access the main tutor workspace dashboards.

Step 7: Explore topic explanations, chat with the AI tutor, or generate customizable mock quizzes.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	AI responses depend on Google Gemini API availability	Works when API is active and internet connection is available
2	Quiz generation speed may vary based on Gemini API response latency	A retry mechanism is implemented in backend to handle rate limits
3	Currently runs locally; not deployed publicly	Deployment configurations (Dockerfile/Render) are planned for next release